

REGULATORY DOSSIER SUBMISSION

Dossier ID: DOS-BC10D1E76B

Country: Burkina Faso

Submission Date: 2025-12-03

Product: Famozen (famotidine) | ATC: A02BA03 | Strength: 20 mg

Applicant: National Therapeutics | Manufacturer: Botswana Clinical Manufacturing

Policy Label: reject_and_return | Risk Score: 0.92

AMR Stewardship: AWaRe=not_applicable | Unmet Need=not_applicable | GLASS Trend=not_applicable
| Watch Similarity=not_applicable

SECTION CONTENT

[1] m1_application_admin - Application Form and Administrative Information

Labels: presence=present; length=length_ok; correctness=correct

This dossier constitutes a formal application for Marketing Authorization of Famozen (famotidine) submitted on 2025-12-03 to the national regulatory authority of Burkina Faso. The applicant, National Therapeutics, formally requests approval for the commercial distribution of this medicinal product. The proposed pharmaceutical form is tablet with a strength of 20 mg, classified under ATC code A02BA03. All required administrative declarations, legal attestations, letters of authorization, and regional application forms have been completed, signed by the responsible Qualified Person, and appended to Module 1. Regional administrative forms were checked for signature completeness and dossier version alignment. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The applicant confirms alignment with applicable technical guidance and regional submission standards. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Applicant authorization documents and legal attestations were reconciled against the named manufacturing parties. A comprehensive table of contents and glossary of abbreviations are provided to facilitate regulatory review. All referenced documents are available in the dossier annex and traceable by section identifier. Where applicable, protocol deviations were assessed for impact on interpretation

[1] m1_manufacturer_gmp - Manufacturer and GMP Evidence

Labels: presence=present; length=length_ok; correctness=incorrect

Error tags: gmp_certificate_expired

Primary manufacturing site: Botswana Clinical Manufacturing in Botswana. Latest GMP inspection date: 2024-12-14. GMP status: compliant. GMP certificate number: GMP-757193; certificate expiry: 2024-08-31. Certificate has expired and requires renewal before authorization. Method validation records and quality controls were reviewed for internal consistency. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The applicant confirms alignment with applicable technical guidance and regional submission standards. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating

acceptable accuracy, precision, and specificity. All referenced documents are available in the dossier annex and traceable by section identifier. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Batch manufacturing records and standard operating procedures (SOPs) are maintain

[1] m1_product_information - Product Information and Naming

Labels: presence=present; length=length_ok; correctness=correct

The proposed proprietary name for this medicinal product is Famozen, containing the active pharmaceutical ingredient famotidine. The draft Summary of Product Characteristics (SmPC), Patient Information Leaflet (PIL), and primary/secondary packaging labels are provided for review. These documents detail the approved therapeutic indications, posology, contraindications, special warnings, and precautions for use. A comprehensive risk management plan to minimize medication errors is also submitted. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The manufacturing site operates under a fully implemented Pharmaceu

[2] m2_quality_overall_summary - Quality Overall Summary

Labels: presence=present; length=length_ok; correctness=correct

This Quality Overall Summary (QOS) provides a critical evaluation of the chemistry, manufacturing, and controls (CMC) data for famotidine. The control strategy justifies the proposed specifications for both the active substance and the finished product, emphasizing critical quality attributes (CQAs) and critical process parameters (CPPs). Impurity profiles, including degradation products and potential genotoxic impurities, have been thoroughly characterized. Batch analysis data from three commercial-scale validation batches demonstrate consistent manufacturing performance and compliance with all release criteria. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. All referenced documents are available in the dossier annex and traceable by section identifier. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The applicant confirms alignment with applicable technical guidance and regional submission standards. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Method validation records and

[2] m2_clinical_overview - Clinical Overview and Benefit-Risk Summary

Labels: presence=present; length=length_ok; correctness=incorrect

Error tags: clinical_failed

Clinical overview for type 2 diabetes indicates pivotal efficacy outcomes did not meet the primary endpoint. Current evidence does not support a favorable benefit-risk conclusion for authorization. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Statistical significance was set at an alpha level of 0.05, and all tests were

two-sided. Clinical interpretation focuses on whether efficacy, safety, and benefit-risk conclusions are supported by the submitted evidence. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Method validation records and quality controls were reviewed for internal consistency. Benefit-risk language was reviewed for consistency with the submitted efficacy and safety summaries. All referenced documents are available in the dossier annex and traceable by section identifier. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The applicant confirms alignment with applicable technical guidance and regional submission standards. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The clinical overview links the therapeutic rationale to the pivotal evidence package and the proposed indication. The target population demographics are broadly representative of the intended commercial patient cohort. No new safety signals were identified during the extended open-label follow-up period.

[3] m3_api_quality - API Quality and Control Strategy

Labels: presence=present; length=length_ok; correctness=correct

The active pharmaceutical ingredient (API) section details the complete synthesis route, starting from well-defined regulatory starting materials. Robust in-process controls and intermediate specifications ensure the consistent quality of the final drug substance. The analytical procedures used for release and stability testing have been fully validated for accuracy, precision, linearity, and robustness. The impurity control strategy adequately addresses organic impurities, residual solvents, and elemental impurities in accordance with current ICH guidelines. Forced degradation studies confirm the stability-indicating nature of the assay methods. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The API package links specification limits to validated analytical methods and impurity-control decisions. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Critical material attributes and control strategy elements were reviewed for consistency across the quality dossier. The synthetic route involves three well-characterized steps with defined starting materials and isolated intermediates. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. All referenced documents are available in the dossier annex and traceable by section identifier. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The applicant confirms alignment with applicable technical guidance and regional submission standards. Batch analysis, validation, and impurity management records support the proposed API quality framework. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Method validation records and quality controls were reviewed for internal consistency. The container

closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline

[3] m3_fpp_manufacturing - FPP Manufacturing Process and Controls

Labels: presence=missing; length=missing; correctness=incorrect

Error tags: missing_critical_section

[MISSING SECTION]

[3] m3_stability - Stability and Shelf-Life Justification

Labels: presence=present; length=length_ok; correctness=correct

A comprehensive stability program has been executed to justify the proposed shelf-life and storage conditions. Data from both accelerated (40°C/75% RH for 6 months) and long-term storage conditions (spanning up to 36 months) are presented for multiple primary stability batches. Statistical evaluation using linear regression and poolability tests confirms that degradation trends remain well within the acceptable specification limits. Photostability and freeze-thaw cycling studies demonstrate that the product does not require specific handling precautions. A post-approval stability protocol commitment is included. Method validation records and quality controls were reviewed for internal consistency. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The applicant confirms alignment with applicable technical guidance and regional submission standards. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. All referenced documents are available in the dossier annex and traceable by section identifier. The risk management plan includes routine pharmacovigilance and proposed educ

[4] m4_nonclinical_summary - Nonclinical Study Summary

Labels: presence=present; length=length_ok; correctness=correct

The nonclinical testing program evaluated the primary pharmacodynamics, secondary pharmacodynamics, safety pharmacology, and comprehensive toxicology profile of the compound. Repeat-dose toxicity studies in two mammalian species established clear no-observed-adverse-effect levels (NOAELs). Genotoxicity testing (Ames and in vivo micronucleus assays) yielded negative results. Reproductive and developmental toxicity studies revealed no teratogenic signals at clinically relevant exposures. The calculated safety margins support the proposed clinical dosing regimen without major safety concerns. The applicant confirms alignment with applicable technical guidance and regional submission standards. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. All referenced documents are available in the dossier annex and traceable by section identifier. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating

acceptable accuracy, precision, and specificity. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The container closure system consists of PVC/PVDC-Aluminum blisters, comp

[5] m5_trial_listing - Tabular Listing of Clinical Studies

Labels: presence=present; length=length_ok; correctness=correct

The tabular listing of clinical studies provides a comprehensive inventory of all trials conducted in support of the proposed indication (type 2 diabetes). This includes Phase I pharmacokinetic/pharmacodynamic studies, Phase II dose-finding studies, and the pivotal Phase III efficacy and safety trials. For each study, the table details the protocol identifier, study design, randomization ratio, number of enrolled subjects, study duration, and current completion status. The applicant confirms alignment with applicable technical guidance and regional submission standards. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Method validation records and quality controls were reviewed for internal consistency. Adverse events were coded using the Medi

[5] m5_pivotal_trial_reports - Pivotal Clinical Trial Reports

Labels: presence=present; length=length_ok; correctness=incorrect

Error tags: clinical_failed

Pivotal trial analysis indicates primary endpoint was not met in the intent-to-treat population. Estimated treatment effect did not reach pre-specified significance and sensitivity analyses were inconsistent. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Method validation records and quality controls were reviewed for internal consistency. The trial reports connect efficacy conclusions to the prespecified analysis framework and observed safety findings. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Discontinuations due to treatment-emergent adverse events (TEAEs) were low and balanced across all study arms. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The pivotal study narratives align endpoint definitions, analysis populations, and outcome interpretation. The applicant confirms alignment with applicable technical guidance and regional submission standards. All referenced documents are available in the dossier annex and traceable by section identifier. The primary efficacy endpoint was met, demonstrating a statistically significant improvement over placebo ($p < 0.001$). The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Clinical study reports document protocol adherence, endpoint handling, and safety interpretation in the target population. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Data were analyzed using

a mixed-effects model for repeated measures, adjusting for baseline covariates. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The manufactu

[5] m5_bioequivalence - Biopharmaceutics and Bioequivalence Evidence

Labels: presence=present; length=length_ok; correctness=correct

Biopharmaceutics data substantiate the formulation development bridging between clinical trial materials and the proposed commercial product. In vivo bioequivalence studies demonstrated that the 90% confidence intervals for the geometric mean ratios of Cmax and AUC parameters fell entirely within the standard acceptance criteria of 80.00% to 125.00%. Additionally, multi-point comparative dissolution profiles across physiological pH ranges (pH 1.2, 4.5, and 6.8) confirmed similarity ($f_2 > 50$). The bioanalytical methods used for plasma quantification were fully validated. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Method validation records and quality controls were reviewed for internal consistency. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Stability testing was conducted in climatic zones III an